PDD - BYOC SIP Trunk Monitoring

Table of Contents:	**Feature Overview**	
	WIP Feature name	-
	Product Manager	@Sreekanth Narayanan
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	Feature(s) on Jira	TDI-6496: BYOC SIP Trunk Monitoring & Reporting - Explore BACKLOG
	Technical Design Definition	[TDD] BYOC Monitoring
	Product Design Artifacts	-

Executive Summary

Purpose of the Product or Feature	By adding BYOC monitoring, we will address this gap and be able to close the highest CREQ for GCN (1385). This requirement is also listed as a Top 10 CREQ in the Critical CR list. CREQ-342: SIP Trunk Monitoring and Reporting IN ROADMAP Bring Your Own Carrier allows Talkdesk users to connect SIP trunks of their choice to their Talkdesk instance, including connections to carriers and third-party telephony platforms. However, no metrics for these connections are available in the Talkdesk UI or for consumption via API, which leaves users blind as to the state and usage of the SIP trunk. **Note**: BYOC in this context relates to Talkdesk customer BYOC trunks. It does not consider TD Minutes carriers connected via xConnect (e.g. BICS, Inteliquent, etc.).

Why are we building this, and why now?	This functionality is considered table stakes. Many of our competitors provide this functionality as well. Not providing this functionality gives the impression that we haven’t covered some of the basic features that should be present.
Total Addressable Market	All Talkdesk customers that use BYOC will have access to this feature.
Keywords	• BYOC: Bring Your Own Carrier. A feature that allows a Talkdesk customer to connect one or more SIP trunks between a carrier or third-party telephony platform and their Talkdesk instance. • SIP Trunk/Interconnect: A connection between Talkdesk and a third-party over which voice traffic traverses. • xCAT: xConnect Administration Tool. A tool used to configure SIP trunks connecting to the Talkdesk platform. xCAT is available internally (GCN & PS) as of November 2023. This tool is *not* available externally. • xConnect: Collective name for infrastructure components that enable the connection of SIP trunks and the BYOC feature.
Target Audience	Administrators, Partners, Resellers
The Job	Add metrics for SIP trunk statuses as well as usage, which can be consumed via APIs as well as in the Talkdesk UI.
The Pain	Administrators of Talkdesk organizations or their customer’s organizations today do not know if the SIP trunk between TD and their enterprise or carrier is online. If there is degraded service or an outage, their actions are reactive, i.e. the problem has already occurred and they come to know about it through a ticket from their users. They also do not have ways of monitoring the trunk usage, which could provide some useful statistics to them (e.g. call volume, CPS, call quality, trends).
The Gain	Administrators will be able to realise extra value with BYOC and we will provide a much requested feature, for Customers, Partners and Telco Resellers.

Critical Change	This feature is not a critical change. It adds the ability for administrators to monitor SIP trunks and provides statistics. There are no changes with existing behavior of the desktop or experience.

Scope and Specifications

<table>
<tr>
<td>Core Functionalities</td>
<td>
Since this is a monitoring feature, the core functionalities will be the key metrics that we’d like to track on the SIP trunks. These metrics should be available in both Explore as well as Live API.
<table>
<tr><th>Metrics</th><th>Granularity, Data points</th></tr>
<tr><td>Trunk Status:
This metric represents the operational state of the SIP trunk</td><td>The different statuses are:
Up
Down
Degraded</td></tr>
<tr><td>Concurrent Inbound calls:
This metric represents the total number of inbound calls received by the customer's BYoC SIP Trunk(s) that were active simultaneously during a defined time period.</td><td>Granularity:
5 seconds
Data Points: Call start time, call end time, SIP Trunk ID, origin number, destination number, call duration, call status (connected, failed, busy, etc.).</td></tr>
<tr><td>Concurrent Outbound calls:
This metric represents the total number of outbound calls initiated by the customer's BYoC SIP Trunk(s) that were active simultaneously during a defined time period.</td><td>Same as above</td></tr>
<tr><td>Inbound Calls:
This metric represents the total number of inbound calls</td><td>Granularity:
Aggregated count (e.g., per minute, per 5 minutes, per</td></tr>
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received by the customer's BYoC SIP Trunk(s) that were active during a defined time period.	hour). **Data Points:** Call start time, call end time, SIP Trunk ID, origin number, destination number, call duration, call status (connected, failed, busy, etc.).
Outbound Calls: This metric represents the total number of outbound calls **initiated** by the customer's BYoC SIP Trunk(s) that were active during a defined time period.	**Granularity:** For Live reporting: Per 5 seconds For explore reporting: Aggregated count (e.g., per minute, per 5 minutes, per hour). **Data Points:** Call start time, call end time, SIP Trunk ID, origin number, destination number, call duration, call status (connected, failed, busy, etc.).
Trunk Status: The current operational state of each configured SIP Trunk. This indicates whether the trunk is registered, active, experiencing issues, or down.	**Granularity:** Real-time (last known status, with updates as changes occur). **Data Points:** Trunk ID, current status (e.g., "Active", "Inactive", "Degraded", "Unregistered"), last status change timestamp. **Alerting:** Critical for immediate notification of trunk downtime or degradation.
Rolling MOS Score (Mean Opinion Score): A numerical measure of the perceived quality of a voice	**Granularity:** Per-call calculation, aggregated average over a rolling time

	call, typically on a scale of 1 (poor) to 5 (excellent). "Rolling" implies a continuous average over a recent time window (e.g., last 5 minutes, last 10 calls).	window (e.g., 5-minute average, 15-minute average). **Data Points:** MOS score per call, aggregated average MOS score per trunk, minimum/maximum MOS score in period. **Thresholds (Reference):** • 4.0 - 5.0: Excellent quality • 3.5 - 4.0: Good quality • 2.5 - 3.5: Fair quality (issues noticeable) • 1.0 - 2.5: Poor quality (unacceptable)`

	RTT:
	
	The time taken by a packet to reach from source to destination and back.
	
	Measurement: Milliseconds (ms)
		Granularity:
		
		• **Live:** Average RTT per call, aggregated average RTT over a rolling time window (e.g., 5-minute average, 15-minute average per trunk).
		• **Explore:** Historical trend of average RTT over specified time ranges (hourly, daily, weekly), peak and distribution of RTT values.
		• **Thresholds (Reference):**
		  ◦ Acceptable: Below 150ms
		    ▪ Problematic: Above 250ms

	Jitter: The variation in the delay of received packets, which can cause choppy or distorted audio.	**Granularity:** • **Live:** Average jitter per call, aggregated average jitter over a rolling time window (e.g., 5-minute average, 15-minute average per trunk).

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<tr>
<td>Measurement: Milliseconds (ms).</td>
<td>

Explore: Historical trend of average jitter over specified time ranges (hourly, daily, weekly), peak jitter values, distribution of jitter values.
Thresholds (Reference):

Acceptable: Below 30ms (preferably below 20ms)

Problematic: Above 30ms

</td>
</tr>
<tr>
<td>Packet Loss:

The percentage of data packets that fail to reach their intended destination during transmission, leading to missing audio or communication breakdowns.</td>
<td>Measurement: Percentage (%).

Granularity:

Live: Packet loss percentage per call, aggregated average packet loss over a rolling time window (e.g., 5-minute average, 15-minute average per trunk).
Explore: Historical trend of packet loss over specified time ranges, peak packet loss percentages, distribution of packet loss.
Thresholds (Reference):

Acceptable: Less than 1%
Problematic: Above 1%

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<tr>
<td>Non-Core Functionalities</td>
<td>Historical data for metrics

Number of degraded or trunk down incidents over the last 6 months or year
MOS and other stats graphs over the last 6-12 months.

Trends and pattern recognition in the metrics to provide analysis. Administrators and NOC teams should also be able to</td>
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	• Observe occurrences of issues over a period of time, such as the last 6-12 months • Predictions based on historical data, such as degraded service or outages occurring in specific time ranges or days of the year
Roles & Permissions	No additional permissions or roles needed.
Out of Scope	BYOC in this context relates to Talkdesk customer BYOC trunks. It does not consider TD Minutes carriers connected via xConnect (e.g. BICS, Inteliquent, etc.).
Assumptions & Constraints	**Assumptions** • Metrics can be obtained from xConnect infrastructure components. **Constraints** • The listed metrics may not be available from the xConnect infrastructure. • Additional code and modules may be needed in order to send the metrics required in this feature.
Limitations	• Real-time metrics are likely not feasible. As such, we are aiming for <=5 second updates.
Prerequisites & Dependencies	**Prerequisites** xConnect has the necessary plugins and software to emit the metrics. **Dependencies** Consume metrics from xConnect - Reporting (Data Platform) Populate and expose dataset for consumption by customers in Live and Explore, including via API - Reporting (Data Platform)

User Experience & Design

- User Stories

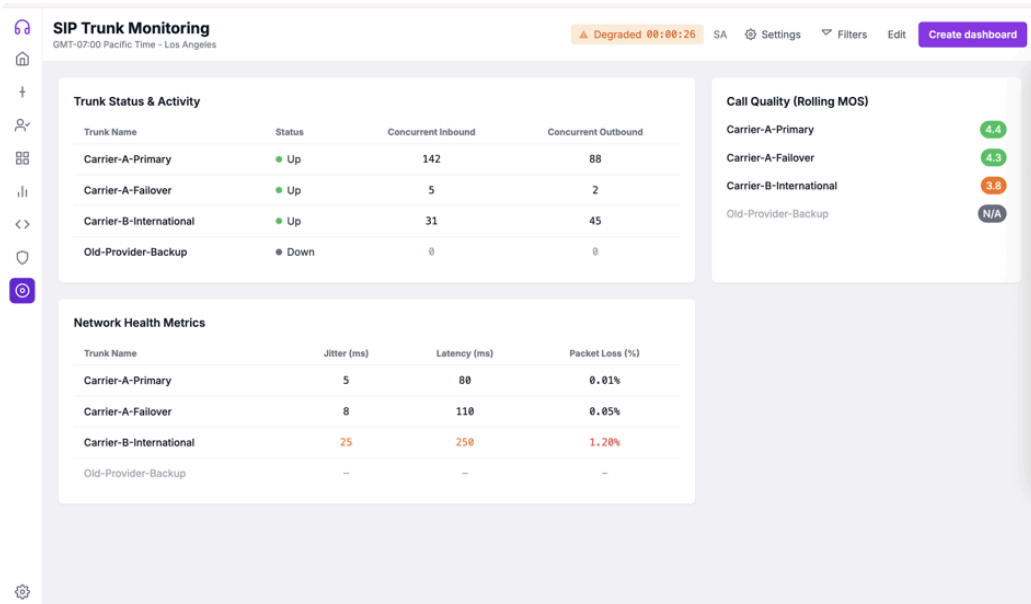
User Story	Description	MoSCoW
As an administrator, I want to have available core metrics for • Trunk name • Trunk status (up/down) • Concurrent inbound calls • Concurrent outbound calls • Rolling Mean Opinion Score (MOS) so that I can determine the state and use of a trunk at any time.	Core metrics are those that are considered MVP for this feature. Metrics must be available per trunk, noting that each Talkdesk instance may have more than one trunk configured. MOS score helps determine the quality of the audio traversing the trunk. Quality is one of the basic metrics that an administrator should be able to observe.	MUST HAVE
As an administrator, I want the core metrics to be available in near real-time so that I know the statistics are up to date.	Near real-time is considered <= 20 seconds.	MUST HAVE
As a developer, I want to be able to consume core BYOC metrics from the Live and Explore API, so that I can ingest data in external applications.	Customers expect to be able to subscribe to a live (near real-time) feed and request metrics on demand. Developers in customer organizations build customized dashboards by calling APIs and listening to events from Talkdesk.	MUST HAVE
As an administrator or user, I want to consume core BYOC metrics from the Live and Explore reporting tools in Talkdesk so that I can create	We should provide administrators and users the ability to create dashboards in Talkdesk Live as well as	MUST HAVE

dashboards and reports based on the BYOC monitoring dataset.	Explore to monitor BYOC metrics within the Talkdesk UI.	
As an administrator, I want to see enhanced metrics for • *Trunk friendly name • Jitter • Latency • Packet loss so that I can report on my trunks more accurately.	Enhanced metrics provide more value and better insight into how a trunk is performing. *Trunk friendly name is dependent on the feature being released in xCAT.	SHOULD HAVE
As an administrator, I want the enhanced metrics to be available in near real-time so that I know the statistics are up to date.	Near real-time is considered <=5 seconds.	SHOULD HAVE
As a developer, I want to be able to consume enhanced BYOC metrics from the Live and Explore API, so that I can ingest data in external applications.	Customers expect to be able to subscribe to a live (near real-time) feed and request metrics on demand. Developers in customer organizations build customized dashboards by calling APIs and listening to events from Talkdesk.	SHOULD HAVE
As an administrator or user, I want to consume enhanced BYOC metrics from the Live and Explore reporting tools in Talkdesk so that I can create dashboards and reports based on the BYOC monitoring dataset.	We should provide administrators and users the ability to create dashboards in Talkdesk Live as well as Explore to monitor BYOC metrics within the Talkdesk UI.	SHOULD HAVE

Mockups

The mockups below give an administrator a quick view in to all the essential information needed for each trunk that they currently have.

This is a mockup of the Live dashboard created automatically in the Talkdesk UI. The above look can be customized by administrators / teams as needed. They can also for instance, change the “Service Status” widget to show a list of the current trunks and their statuses.



The second mockup is generated. It may not reflect exactly how data is visualized in the Live dashboard.

In the Explore section, all of this could simply be visualized as a table with each SIP Trunk name being the unique key, while the remaining parameters become the values.

Customer journeys

Consumption of metrics by external platform via API

An administrator or the NOC team uses a NOC monitoring application for their infrastructure, centralising visibility of their platforms, data and voice connections. They would like to include Talkdesk BYOC SIP trunk in their monitoring. Therefore, the next step is for the developers to modify the NOC application to subscribe to the Live API for SIP trunk metrics. Once subscribed, the subscription pushes updates in near real-time. The NOC monitoring application consumes the data and provides a visual representation. From the data received, the NOC team configures alerts based on their required thresholds.

Create Live dashboard on Talkdesk UI

An administrator or user wishes to view SIP trunk metrics in the Talkdesk UI. They open Live reporting and creates a new dashboard. Then, the user adds metrics from the SIP trunk dataset. Once the graphs are in place, the user can add alarm thresholds and visualisation options as required.

Technical Requirements

Architecture Diagram	This diagram provides a high-level overview of the metrics that would flow from X-Connects across various geographies. The X-Connect consumer has code to collect CDR data from the X-Connect clusters. It also contains data classes for CDR which seems to contain fields such as jitter, latency. Additional code might be required to structure the data as needed at the SIP trunk level. How to integrate with Data Platform 2.0 The X-Connect consumer should send data to Morozko, which would then classify the data into Live and Historical and send the data to AWS RDB & Iceberg respectively. Additional details are available in the Data Platform document linked.
3rd Party Vendors	No 3rd party vendors
Security & Compliance	We will not record any PII in these metrics. Therefore, there is no impact to security and compliance.

Dependencies	List known dependencies on other Talkdesk teams or third parties including partners and customers: • Reporting (Data Platform) Team: we will need to work on the Live dashboards with them, the look and feel as well as what needs to be displayed.
Viability of building with AI	Using AI to build this feature has limited potential. Reasons include: • This feature requires context about existing solution architecture and would involve changes across multiple components. Due to this spread, an AI tool will not be able to provide appropriate change-sets across all components. • We aren't building a new microservice, so new tooling and completely new code may not be needed. • The metrics for monitoring SIP calls will need to be analyzed and dashboards created accordingly for Explore and Live. These dashboards will need to be created manually.

Competitor Analysis

For whom are we creating value? Who are our most important customers? Is this aligned with an industry opportunity?

Do competitors have something similar? What differentiates us? See the table below.

Product/Feature	Genesys	Cisco	Five9	Nice	Vonage	Avaya
SIP Trunk/Edge monitoring	Genesys Cloud Monitoring through App Foundry It looks like this service can be purchased	A customer would have to use theh Voice PoP onboarding guide and install Cisco CUBE to connect to their own	Doesn't support monitoring the BYOC trunk. Five9 BYO SIP Trunk document states in	There doesn't seem to be any documentation around configuring a SIP trunk for BYOC scenarios.	Diagnostics is located here: Vonage API Developer Found a video that shows the reporting of calls on	Avaya SBCs provide ways to monitor and log them using SNMP.

	by the customer through a partner.	carrier or PBX. Peers on CUBE can be monitored using SNMP.	section 9.3 that *Five9 will not monitor the customer's endpoints/sbcs/pbxs. Five9 does not support path checks to customers, but Five9 does respond to option pings destined to Five9 signaling IPs sourced from known customer IP addresses.*	Not sure about the monitoring aspect as well.	Vonage, but it doesn't look like there is a dashboard. [Vonage Reporting & Analytics Demo]()	

Roadmap & Timeline

Release Milestones & Timeline	To manage the complexity and ensure timely delivery using Agile methodology, we recommend that the project should be split into 4 development phases. When an issue occurs in the field, an administrator or NOC team generally wants to figure out the status of the trunk ASAP. Based on the purposes of Explore and Live, it makes more sense to display the data on Live first followed by Explore.

Phase 1: Core Metrics in Live (Dashboard Widget)

Objective: Implement core metrics in the Talkdesk Live dashboard widget. Ex.: https://ucprod.mytalkdesk.com/atlas/apps/live

Features:

- Integrate basic SIP trunk status and usage metrics into a real-time dashboard widget.
- Create visual indicators and charts for core metrics in the dashboard.

Phase 2: Enhanced Metrics in Live (Dashboard Widget)

Objective: Enhance the dashboard widget with Enhanced metrics. Ex.: https://ucprod.mytalkdesk.com/atlas/apps/live

Features:

- Integrate detailed call statistics and call quality metrics into the real-time dashboard widget.
- Create advanced visual indicators, charts, and alerts for enhanced metrics in the dashboard.

Phase 3: Core Metrics in Explorer (Looker)

Objective: Implement core metrics for SIP trunk status and usage in Explorer (Looker). Ex.: https://ucprod.mytalkdesk.com/atlas/apps/explore

Features:

- Collect and store basic SIP trunk status (up/down).
- Collect and store basic call statistics (inbound/outbound concurrent calls) from SBC CDRs.
- Integrate core metrics into Looker for user exploration and visualization.

Phase 4: Enhanced Metrics in Explorer (Looker)

Objective: Implement Enhanced metrics available in Explorer (Looker) with advanced data. Ex.: https://ucprod.mytalkdesk.com/atlas/apps/explore

Features:

- Collect and store detailed call statistics from SBC CDRs.

	• Collect and store call quality metrics from the RTP engine. • Integrate enhanced metrics into Looker for advanced user exploration and visualization.
Risks	• The listed metrics may not be available from the xConnect infrastructure. In this case, we will need to add more components or code to emit the required set of metrics, or have to approximate certain statistics from other sources like CDRs.

Go-To-Market Strategy & Enablement

PMs to host regular live enablement sessions (walk-throughs, demos, Q&A) for GTM SMEs.

Rollout Plan	• We should do a progressive rollout of the metrics emission in various xConnect clusters. The rollout can be done across 2 weeks.
	  ◦ Week 1 - APAC. We monitor the xConnect infrastructure and ensure there are no impact to calls. The lower volume in APAC will also give us a way to quickly verify if everything is working well on a smaller set of calls.
	  ◦ Week 2 - EU. We rollout the changes to xConnect infrastructure in Europe and monitor.
	  ◦ Week 3 - US & Canada. This would be the largest chunk of calls. After 2 weeks of monitoring existing metrics, we can be confident of nothing untoward happening, and enable the metrics for this geography too.
	  ◦ Week 4 - Any other clusters needed for full deployment.
	• There are no 3rd parties involved in the rollout. There is no new licensing or SKU.

Preview to GA	• It would be good to set up a preview for a subset of customers willing to test this feature out and provide feedback to us at an early stage. • The CREQ in Salesforce has a list of Risks associated with it (17 in total). The product team can reach out to CSMs for the respective risks and make a plan for the preview.

	• The preview phase can be 30 days where we'd share mockups with the chosen candidates and get the requisite feedback.
Provisioning	• We will not need any customer information to configure the product/feature. This applies to all customers. • No setup needed from a user or customer. We could provide a guide to build the dashboard or report and show it as a demo. • No steps needed to activate the reporting. The ability to use Explore or Live show be available as soon as we start to emit the required metrics from xConnect.
Configuration	-
Support & Troubleshooting	Describe troubleshooting and common errors for both customers and internal Talkdesk Support Representatives. Please add the link to the supporting documentation when available.
Pricing & Monetization More info here	• This feature will not be monetized. There are no SKUs for it.

Success Metrics & KPIs

KPIs	Describe how success will be measured for this feature. Adoption, retention, usage, revenue, etc Success of this feature will be measured using the following statistics: 1. Number of customers using the Trunk dashboard on Talkdesk Live monthly 2. Number of reports downloaded monthly 3. Frequency of reports being downloaded/viewed monthly
Customer Feedback Mechanisms	Surveys, reviews, preview Pilot program with a few customers to get early feedback on the feature.

Appendix

References: Include here any relevant documents, DoD, FAQs, research, product team slack, etc.